

- Q1. Acyclic graph - A graph without cycles
- Q2. Connected graph - every vertex is reachable from every other vertex
- Q3. Adjacency matrix - uses a 2D array
- Q4. Directed graph - model one-way roads
- Q5. Directed graph - models one-way roads
- Q6. Tree - A connected acyclic graph
- Q7. Directed Acyclic Graph (DAG) -
- Q8. Property required before topological sorting: Graph must be acyclic
- Q9. Graph commonly used in job assignment: Bipartite graph
- Q10. Graph representing hierarchical relationship: Tree
- Q11. Directed Acyclic Graph (DAG) - topological ordering exists
- Q12. Unprocessed vertices after topological sorting indicate: A cycle exists
- Q13. Data Structure for DFS-based topological sorting: Stack
- Q14. Technique for detecting cycle in topological sorting: DFS
- Q15. Graph model suitable for course prerequisite system: DAG
- Q16. Data Structure improving node selection in Kahn's algorithm: Queue
- Q17. Graph used in compiler build systems: DAG

- Q18. MST algorithm selecting edges from a starting vertex
Prim's algorithm
- Q19. Shortest Path algorithm that detects negative cycle: Bellman - Ford algorithm
- Q20. Traversal for shortest paths in unweighted graphs: BFS
- Q21. Repeated operation in shortest path algorithms: Relaxation
- Q22. Algorithm using a greedy strategy for shortest path: Dijkstra's algorithm
- Q23. Algorithm using dynamic programming for shortest paths: Floyd - warshall algorithm
- Q24. Time complexity of Floyd-warshall: $O(V^3)$.
- Q25. Data structure used in Floyd-warshall: 2D Distance matrix
- Q26. Traversal relying on recursion or stack: DFS
- Q27. Process of visiting all vertices: Graph traversal
- Q28. Concept applied in network design optimization: Minimum Spanning tree (MST)
- Q29. Removing the heaviest edge from a cycle ensures Minimum Spanning tree property.
- Q30. Property guaranteeing correctness of an MST: Cut Property
- Q31. Algorithm computing all-pairs shortest path: Floyd-warshall

- Q32. Core concept behind repeated shortest - path updates: relaxation
- Q33. Traversal to check whether a path exists: BFS or DFS
- Q34. Strategy used in uniform - cost search: greedy selection of lowest - cost node.
- Q35. Traversal guaranteeing shortest path in unweighted graph: BFS
- Q36. Another term for BFS: Breadth First Search
- Q37. Traversal approach used in Kahn's algorithm: BFS-like traversal
- Q38. topological sorting method using recursion: DFS
- Q39. Sequence of vertices from source to destination: path
- Q40. Graph having weighted edges: weighted graph.